Lesson 8.1: System prompt optimisation, context window management and artifacts

What you learn

- **System prompts.** State the role, the audience, rules, format and what to do when unsure. Test with hard cases and keep versions.
- **Context window.** A model can read only so much text at once. Keep instructions, the relevant documents and recent turns; trim or summarise the rest (see the tested code).
- **Artifacts.** In Claude, artifacts are separate documents, pages or small apps produced beside the chat, which you can edit and share.
- **Evaluation.** Write 20 test questions and compare prompt versions on them.

Tools and resources (official sites, not checked live)

- Anthropic prompt engineering
- Claude

ASSIGNMENT

Practice for lesson 8.1

Business problem: Trim a long chat.

Steps

1. Run the tested trimmer.
2. Change the budget.
3. See which messages remain.

Expected output: The system message and newest messages that fit.

SOLUTION (ANSWER KEY)

PYTHON

```
def rough_tokens(text): return max(1, len(text) // 4)   # rule of thumb: about 4 characters
per token
def trim_history(messages, budget):
    """Keep the system message and the newest messages that fit inside the token budget."""
    system, rest = messages[0], messages[1:]
    kept, used = [], rough_tokens(system["content"])
    for m in reversed(rest):
        t = rough_tokens(m["content"])
        if used + t > budget: break
        kept.append(m); used += t
    return [system] + list(reversed(kept))
```

Real tools count tokens exactly. The 4-characters rule is only an estimate.

Check yourself

1. What should always stay in context?
   Answer: The system instructions.
2. How do you compare two system prompts?
   Answer: Run both on the same test questions.

Lesson 8.1 of the **AI and Automation** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.